Comparative Evaluation of Immersive Virtual Reality and Document-Based Training for Industrial Gas Leak Response

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Executive Summary

- Immersive VR training system developed for emergency gas-leak procedures.
- Controlled experiment compares VR simulation to traditional document-based training.
- Evaluated with pre-test, training, post-test, and confidence survey (30 participants).
- VR group showed higher knowledge gains and confidence improvements than document group.
- VR framework offers realistic, risk-free practice and potential for scalable safety training.

Introduction

Safety training in critical industries is of highest priority and must be completed before induction of crew members and staff of such industries. However, industrial safety training has relied only on traditional methods such as technical manuals, machine documentation, and verbal lectures, which fail to set up the right emergency environment and nearperfect replication of a life-threatening situation in an actual emergency.

The 1984 Bhopal gas tragedy, one of the world's worst industrial disasters, is one such example which resulted in thousands of casualties due to lack of emergency response and improper safety training. This motivates us to search for a better training method which shifts the learning paradigm from passive to active . We propose an immersive VR-based guided learning framework for industrial gas leak emergency response and evaluate its effectiveness.

System Design

- Developed in Unreal Engine 4 to simulate an alkylate gas recovery unit (AGRU) environment.
- First-person experience with guided instructions and interactive tasks.
- Spatial audio cues (e.g., gas hissing) for realistic leak detection.
- Visual alarms and blinking lights indicate emergency triggers.
- Interactive tasks: don SCBA, identify leak, perform double-block-and-bleed (DBB) and lockout-tagout (LOTO), operate valves.
- Provides a safe, repeatable training scenario with immersive sensory feedback.

Methodology

- Participants: 30 students (18–24 yrs) with no prior safety experience.
- Two groups: Group A (document-based training), Group B (VR training).
- Pre-test assesses knowledge of SCBA, leak identification, DBB, and LOTO procedures.
- Training: Group A studies written SOP materials; Group B completes the VR simulation.
- Post-test: isomorphic to pre-test to measure learning gains.
- Confidence survey (Likert 1–5) before and after training.
- All participants followed the same experimental flow (see below).

Validated Hypothesis

VR-based training results in higher procedural recall, hazard identification and confidence than Document based training.

Evaluation Metrics

- Knowledge gain: difference between post-test and pre-test scores.
- Improvement (%): (post-test − pre-test)/pre-test, averaged by group.
- Confidence change: self-reported Likert score difference (pre vs post).
- Additional measures: correct hazard identification and procedure compliance (qualitative).

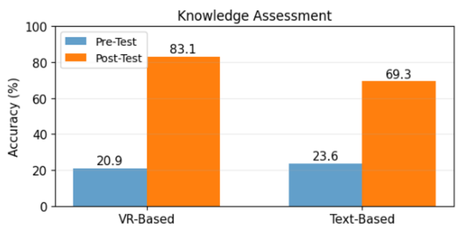


Figure 1:Comparison of average knowledge accuracy and confidence levels before and after training across both experimental groups.

Results

- VR group achieved higher mean post-test score (≈85%) than document group (≈60%).
- Average improvement: VR training ≈35% vs document ≈15%.
- Confidence increased more in VR group (mean +1.5) than in document group (mean +0.8).
- Summary tables and charts illustrate participant demographics and performance.

Metric	Docs	VR
Participants (n)	15	15
Pre-Test Score	3.53	3.13
Post-Test Score	10.40	12.47
Knowledge Gain	6.87	9.43
Post Accuracy (%)	69.33	83.11
Confidence Gain	1.43	2.07
Welch's t-test (p)	$p < 0.01$	

Table 1:SUMMARY OF GROUP-WISE LEARNING, PERFORMANCE AND CONFIDENCE IMPROVEMENT

Conclusion

- This study confirms that VR-based training significantly enhances understanding of safety procedures and increases learner confidence over document instruction.
- Immersive training bridges theory and practice, potentially reducing training costs and errors in real emergencies.
- The proposed framework supports standardized, repeatable drills and performance tracking.
- Limitations include participant diversity and scenario scope; real-world deployment is needed.
- **This work evaluates the effectiveness of immersive VR-based training in improving comprehension of safety-critical procedures compared to traditional document-based instruction.**

Technical Contribution

- VR-guided SOP training model for industrial gas-leak emergencies.
- Integration of spatial audio and interactive DBB/LOTO tasks in Unreal Engine.
- Controlled evaluation framework for comparing training modalities.
- Empirical evidence of improved learning outcomes with VR training.

Future Work

- Incorporate haptic feedback and adaptive scenario difficulty.
- Develop multi-user collaborative emergency drills.
- Conduct field tests with experienced operators.
- Extend system to other hazard scenarios (e.g., fire, spills).